

## ST LEONARD, NB, Canada

WMO: 717030

Lat: 47.158N

Lon: 67.832W

Elev: 242

StdP: 98.45

Time Zone: -4.00 (NAA)

Period: 94-17

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |             |              | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|-------------|--------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%          |              |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS          | MCDB         | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-26.4 | (c)<br>-23.8 | (d)<br>-30.0                  | (e)<br>0.2 | (f)<br>-26.0 | (g)<br>-27.5 | (h)<br>0.3 | (i)<br>-23.4 | (j)<br>12.2           | (k)<br>-10.7 | (l)<br>10.8 | (m)<br>-10.4 | (n)<br>2.6            | (o)<br>300 | (p)<br>0.612 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>10.7            | (c)<br>28.7     | (d)<br>20.1 | (e)<br>26.7 | (f)<br>18.9 | (g)<br>25.1 | (h)<br>18.1 | (i)<br>21.7         | (j)<br>26.0 | (k)<br>20.7 | (l)<br>24.5 | (m)<br>19.7 | (n)<br>23.1 | (o)<br>4.1           | (p)<br>260 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.3                            | 15.4 | 23.7 | 19.3 | 14.5 | 22.7 | 18.4 | 13.7 | 21.6 | 64.6          | 25.7 | 60.9 | 24.7 | 57.3 | 23.1 | 26.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.0              | 8.6  | 7.5 | -31.1                      | 31.5 | 2.9                | 1.3 | -33.2                                              | 32.4 | -34.9      | 33.2 | -36.6      | 33.9 | -38.7      | 34.8 |
|                   |      |     | -31.1                      | 23.4 | 3.0                | 1.2 | -33.3                                              | 24.2 | -35.0      | 24.9 | -36.7      | 25.6 | -38.9      | 26.4 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 4.2    | -12.0 | -10.4 | -4.5  | 2.8   | 10.4  | 15.5  | 18.5  | 17.3  | 12.9  | 6.2   | -0.2  | -7.4  |
|                                                                                  | DBStd       | 11.62  | 7.07  | 6.52  | 5.91  | 4.26  | 4.28  | 3.92  | 2.91  | 3.25  | 4.11  | 4.30  | 5.13  | 6.14  |
|                                                                                  | HDD10.0     | 2972   | 682   | 572   | 450   | 219   | 48    | 3     | 0     | 1     | 17    | 133   | 308   | 539   |
|                                                                                  | HDD18.3     | 5265   | 940   | 806   | 708   | 467   | 249   | 102   | 34    | 59    | 170   | 376   | 557   | 797   |
|                                                                                  | CDD10.0     | 839    | 0     | 0     | 1     | 2     | 59    | 169   | 262   | 227   | 103   | 16    | 1     | 0     |
|                                                                                  | CDD18.3     | 91     | 0     | 0     | 0     | 0     | 2     | 18    | 38    | 27    | 6     | 0     | 0     | 0     |
|                                                                                  | CDH23.3     | 799    | 0     | 0     | 0     | 1     | 52    | 178   | 291   | 224   | 51    | 2     | 0     | 0     |
|                                                                                  | CDH26.7     | 165    | 0     | 0     | 0     | 0     | 10    | 41    | 55    | 51    | 9     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 3.1    | 3.4   | 3.6   | 3.6   | 3.5   | 3.2   | 2.9   | 2.6   | 2.3   | 2.6   | 3.1   | 3.3   | 3.3   |
| Precipitation                                                                    | PrecAvg     | 1060   | 76    | 64    | 71    | 74    | 90    | 102   | 110   | 100   | 98    | 104   | 85    | 88    |
|                                                                                  | PrecMax     | 1327   | 134   | 106   | 155   | 133   | 146   | 179   | 194   | 224   | 190   | 233   | 196   | 171   |
|                                                                                  | PrecMin     | 753    | 15    | 19    | 27    | 34    | 43    | 40    | 34    | 47    | 42    | 29    | 17    | 35    |
|                                                                                  | PrecStd     | 129    | 35    | 19    | 32    | 28    | 28    | 34    | 38    | 44    | 39    | 48    | 40    | 32    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 7.4   | 6.7   | 11.6  | 19.6  | 27.9  | 30.2  | 30.5  | 30.4  | 27.8  | 21.9  | 15.6  | 8.0   |
|                                                                                  |             | MCWB   | 6.7   | 5.1   | 6.3   | 11.3  | 16.1  | 20.6  | 21.1  | 20.8  | 21.1  | 15.1  | 14.2  | 7.4   |
|                                                                                  | 2%          | DB     | 3.7   | 3.5   | 7.5   | 15.1  | 24.5  | 27.7  | 28.1  | 28.1  | 24.5  | 17.7  | 11.9  | 4.7   |
|                                                                                  |             | MCWB   | 3.1   | 2.1   | 3.8   | 8.7   | 14.3  | 18.8  | 20.5  | 20.1  | 18.7  | 13.2  | 9.9   | 4.0   |
|                                                                                  | 5%          | DB     | 1.0   | 1.2   | 5.2   | 12.3  | 21.3  | 25.3  | 26.4  | 25.7  | 22.1  | 15.2  | 9.1   | 2.7   |
|                                                                                  |             | MCWB   | 0.2   | 0.2   | 2.4   | 7.0   | 13.1  | 17.6  | 19.1  | 18.8  | 17.7  | 12.3  | 7.5   | 1.8   |
|                                                                                  | 10%         | DB     | -1.6  | -0.5  | 3.4   | 9.7   | 18.7  | 23.2  | 24.8  | 23.9  | 20.1  | 13.2  | 7.0   | 1.0   |
|                                                                                  |             | MCWB   | -2.3  | -1.5  | 1.2   | 5.3   | 12.1  | 16.8  | 18.4  | 18.0  | 16.3  | 10.6  | 5.7   | 0.3   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 6.9   | 5.6   | 8.4   | 13.3  | 18.7  | 22.3  | 23.4  | 22.4  | 22.0  | 16.6  | 14.4  | 7.7   |
|                                                                                  |             | MCDB   | 7.2   | 6.7   | 9.8   | 17.4  | 23.4  | 27.8  | 27.5  | 27.8  | 26.8  | 19.3  | 15.2  | 8.0   |
|                                                                                  | 2%          | WB     | 3.1   | 2.2   | 4.7   | 10.0  | 16.4  | 20.7  | 21.8  | 21.2  | 20.1  | 14.5  | 10.6  | 4.1   |
|                                                                                  |             | MCDB   | 3.6   | 3.0   | 6.3   | 13.1  | 20.6  | 25.4  | 25.6  | 25.2  | 22.9  | 16.7  | 11.4  | 4.6   |
|                                                                                  | 5%          | WB     | 0.2   | 0.3   | 2.8   | 7.7   | 14.7  | 19.3  | 20.7  | 20.2  | 18.6  | 12.8  | 7.8   | 1.9   |
|                                                                                  |             | MCDB   | 0.9   | 1.2   | 4.8   | 11.4  | 19.0  | 23.2  | 24.5  | 23.8  | 20.9  | 15.1  | 8.7   | 2.5   |
|                                                                                  | 10%         | WB     | -2.2  | -1.5  | 1.3   | 5.8   | 13.0  | 18.0  | 19.6  | 19.2  | 17.0  | 10.9  | 5.8   | 0.4   |
|                                                                                  |             | MCDB   | -1.6  | -0.5  | 3.2   | 9.2   | 17.8  | 21.3  | 23.0  | 22.5  | 19.5  | 12.8  | 7.0   | 1.0   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.4   | 10.0  | 9.7   | 9.6   | 11.4  | 11.4  | 10.7  | 11.3  | 11.1  | 9.0   | 7.1   | 7.8   |
|                                                                                  |             | MCDBR  | 11.3  | 9.5   | 10.9  | 14.7  | 16.8  | 14.9  | 13.3  | 13.9  | 13.3  | 12.8  | 9.9   | 8.2   |
|                                                                                  |             | MCWBR  | 11.3  | 8.6   | 8.2   | 9.5   | 8.8   | 7.8   | 6.8   | 7.3   | 8.1   | 8.9   | 8.7   | 8.0   |
|                                                                                  | 5% WB       | MCDBR  | 11.1  | 8.7   | 9.4   | 12.5  | 12.7  | 11.8  | 10.9  | 10.9  | 10.6  | 11.1  | 8.9   | 8.1   |
|                                                                                  |             | MCWBR  | 11.2  | 8.2   | 7.8   | 9.3   | 8.0   | 7.1   | 6.3   | 6.4   | 7.3   | 8.7   | 8.6   | 8.0   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.254 | 0.270 | 0.301 | 0.341 | 0.351 | 0.370 | 0.403 | 0.382 | 0.356 | 0.318 | 0.289 | 0.272 |
|                                                                                  | taud        |        | 2.345 | 2.283 | 2.278 | 2.368 | 2.422 | 2.369 | 2.297 | 2.357 | 2.422 | 2.526 | 2.527 | 2.377 |
|                                                                                  | Ebn at Noon |        | 847   | 903   | 922   | 914   | 913   | 893   | 856   | 861   | 854   | 843   | 806   | 776   |
|                                                                                  | Edh at Noon |        | 71    | 94    | 111   | 113   | 111   | 118   | 125   | 113   | 97    | 75    | 61    | 62    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 1.39  | 2.27  | 3.46  | 4.44  | 5.01  | 5.44  | 5.42  | 4.74  | 3.72  | 2.22  | 1.25  | 0.96  |
|                                                                                  | RadStd      |        | 0.17  | 0.26  | 0.24  | 0.40  | 0.50  | 0.38  | 0.47  | 0.30  | 0.28  | 0.20  | 0.15  | 0.09  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (25 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page